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MATKUL : PENGOLAHAN CIRTA DIGITAL

KELAS : A

Tugas

Buatkan kode yang menjalankan perkalian matrix untuk array 3 dimensi. Kode dikerjakan dalam bentuk cli python.

Jawab

❖ **Code**

```
❖ def input_matrix(nama, baris, kolom):
❖     matrix = []
❖ 
❖     print(f"\nMasukkan elemen matriks {nama}:")
❖     for i in range(baris):
❖         row = []
❖         for j in range(kolom):
❖             nilai = int(input(f"{nama}[{i}][{j}] = "))
❖             row.append(nilai)
❖         matrix.append(row)
❖ 
❖     return matrix
❖ 
❖ def perkalian_matrix(A, B):
❖     baris_A = len(A)
❖     kolom_A = len(A[0])
❖     kolom_B = len(B[0])
❖ 
❖     hasil = []
❖ 
❖     for i in range(baris_A):
❖         row = []
❖         for j in range(kolom_B):
❖             total = 0
❖ 
```

```
            for k in range(kolom_A):
                total += A[i][k] * B[k][j]

            row.append(total)

        hasil.append(row)

    return hasil

def tampilkan_matrix(matrix):
    for row in matrix:
        print(row)

def main():
    print("====================================")
    print("     PERKALIAN MATRIKS 3 DIMENSI")
    print("====================================")

    jumlah = int(input("Masukkan jumlah matriks: "))

    baris_A = int(input("Masukkan jumlah baris matriks A: "))
    kolom_A = int(input("Masukkan jumlah kolom matriks A: "))

    baris_B = int(input("Masukkan jumlah baris matriks B: "))
    kolom_B = int(input("Masukkan jumlah kolom matriks B: "))

    if kolom_A != baris_B:
        print("\nMatriks tidak dapat dikalikan.")
        print("Kolom matriks A harus sama dengan baris matriks B.")
        return

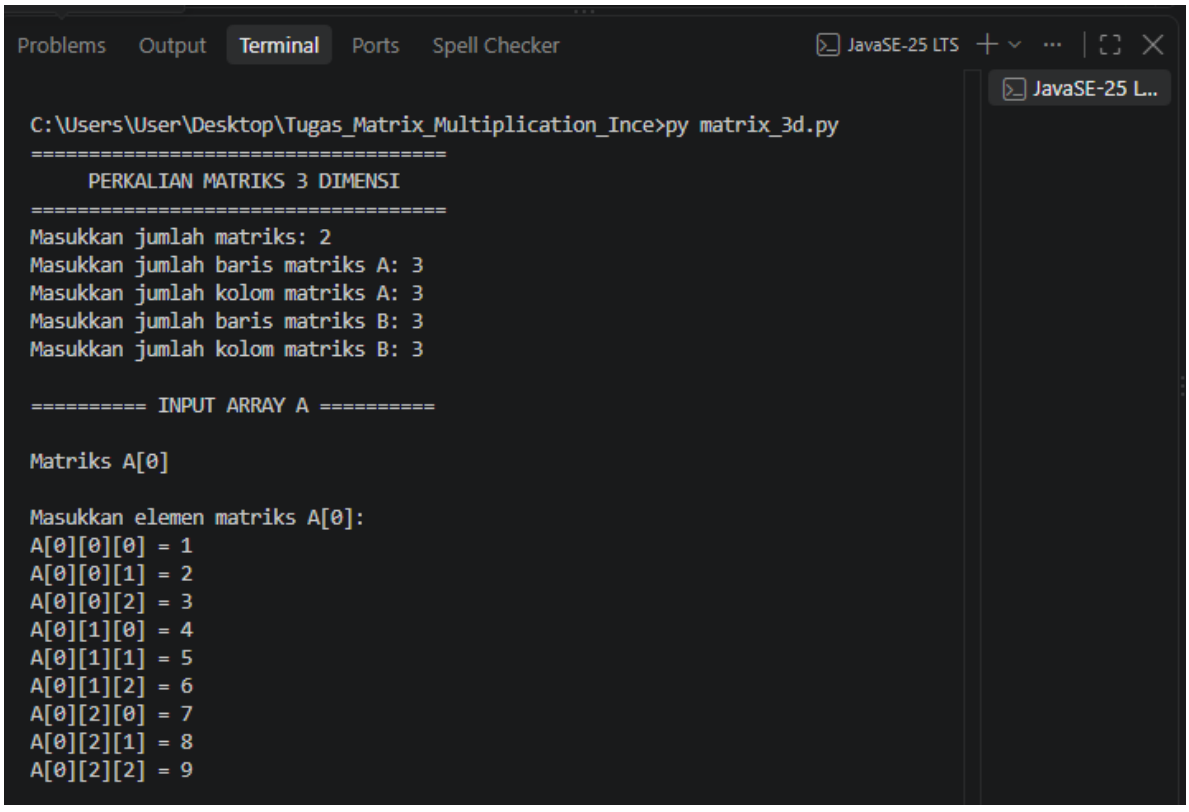
    array_A = []
    array_B = []

    print("\n========== INPUT ARRAY A ==========")
    for i in range(jumlah):
        print(f"\nMatriks A[{i}]")
        A = input_matrix(f"A[{i}]", baris_A, kolom_A)
        array_A.append(A)

    print("\n========== INPUT ARRAY B ==========")
    for i in range(jumlah):
        print(f"\nMatriks B[{i}]")
        B = input_matrix(f"B[{i}]", baris_B, kolom_B)
        array_B.append(B)
```

```
❖
❖      print("\n========== HASIL PERKALIAN ==========")
❖
❖      for i in range(jumlah):
❖          hasil = perkalian_matrix(array_A[i], array_B[i])
❖
❖          print(f"\nHasil matriks ke-{i}:")
❖          tampilkan_matrix(hasil)
❖
❖      print("\n====================================")
❖      print("Perkalian matriks selesai.")
❖      print("====================================")
❖
❖  if __name__ == "__main__":
❖      main()
```

❖ **Output**



```
Matriks A[1]

Masukkan elemen matriks A[1]:
A[1][0][0] = 10
A[1][0][1] = 11
A[1][0][2] = 12
A[1][1][0] = 13
A[1][1][1] = 14
A[1][1][2] = 15
A[1][2][0] = 16
A[1][2][1] = 17
A[1][2][2] = 18
```

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```
========== INPUT ARRAY B ==========

Matriks B[0]

Masukkan elemen matriks B[0]:
B[0][0][0] = 8
B[0][0][1] = 2
B[0][0][2] = 3
B[0][1][0] = 1
B[0][1][1] = 9
B[0][1][2] = 4
B[0][2][0] = 6
B[0][2][1] = 7
B[0][2][2] = 5

Matriks B[1]

Masukkan elemen matriks B[1]:
B[1][0][0] = 20
B[1][0][1] = 21
B[1][0][2] = 22
B[1][1][0] = 23
B[1][1][1] = 24
B[1][1][2] = 25
B[1][2][0] = 26
B[1][2][1] = 27
B[1][2][2] = 28
```

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```
========== HASIL PERKALIAN ==========

Hasil matriks ke-0:
[28, 41, 26]
[73, 95, 62]
[118, 149, 98]

Hasil matriks ke-1:
[765, 798, 831]
[972, 1014, 1056]
[1179, 1230, 1281]

==================================
Perkalian matriks selesai.
==================================
```